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# TA3 – Case-Based Reasoning System

## Case Study 1: Disaster Response Decision Support System

**Purpose.** This program uses previous disaster cases to help suggest a response to a new disaster. The sample scenario is a typhoon with expected flooding in an urban area.

### 1. Case Base

We have 5 sample cases including, Typhoon Ursula, Typhoon Odette, an urban flood, an earthquake, and another typhoon event. Each case contains the disaster type, flooding level, area, severity, affected population, and a previous response. The program stores new cases in a json file called disaster_responses.json. The historical case details are just synthetic examples and does not provide real data from previous actual happenings.

### 2. Similarity Method

The program compares the new event with the saved cases. Disaster type is worth 25 points, flooding level 25 points, area type 20 points, severity similarity 15 points, and population similarity up to 15 points. The case with the highest score is selected as the most similar past event.

### 3. Adaptation

Depending on the severity and level of flooding caused by the new event, the program will recommend an evacuation strategy, actions to take in areas that are prone to flooding, preparation of supplies including food, water, transport, and medical supplies. There will be and estimation on how many people would need to be evacuated, based on 20% of classroom occupants for severe events, 10% for moderate and 5% for low severity.

### 4. CBR Cycle

**Retrieve:** Compare the new event with the stored disaster cases.
**Reuse:** Use the response from the closest past case as a guide.
**Revise:** Adjust the evacuation and resource recommendations for the new event.
**Retain:** Save the new event and its response in disaster_cases.json.

## 5. Sample Program Output

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 DISASTER RESPONSE DECISION SUPPORT
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Number of saved cases: 5

--- NEW DISASTER INFORMATION ---
Event name: Typhoon Test
Disaster type (Typhoon/Flood/Earthquake): Typhoon
Expected flooding (Low/Medium/High): High
Area type (Urban/Coastal/Rural): Urban
Severity (Low/Moderate/Severe): Severe
Estimated affected population: 10000

--- SIMILARITY RESULTS ---
Typhoon Ursula (2019) - 100.0 %
Typhoon Odette (2021) - 52.0 %
Urban Flood Event - 61.5 %
Earthquake Event - 47.0 %
Typhoon Event - 40.0 %

--- MOST SIMILAR PAST CASE ---
Event: Typhoon Ursula (2019)
Similarity: 100.0 %
Previous response: Evacuate residents in flood-prone areas, open shelters, and prepare food, water, and medical supplies.

--- RECOMMENDED RESPONSE ---
Start early evacuation in high-risk communities and help children, older adults, and people who need assistance first.
Focus on low-lying areas and avoid roads that may become flooded.
Prepare transportation, food, drinking water, and basic medical supplies.
Estimated evacuees: 2000
Suggested shelter capacity: 2000 people

New case saved successfully!

--- UPDATED CASE BASE ---
Typhoon Ursula (2019) | Typhoon | High | Urban | Severe
Typhoon Odette (2021) | Typhoon | Medium | Coastal | Severe
Urban Flood Event | Flood | High | Urban | Moderate
Earthquake Event | Earthquake | Low | Urban | Severe
Typhoon Event | Typhoon | Medium | Rural | Moderate
Typhoon Test | Typhoon | High | Urban | Severe

Total cases: 6
```

## 6. Pair Reflection

**CONTRIBUTIONS:**

***PILLONE:***
I worked on the python implementation, including the case retrieval, adaptation, and saving of the new cases being imported to a json file. Development of the case retrieval function to find the most similar disaster based on the scores, recommendations based on the case, etc.

**ONG-IKO:**
I was responsible for organizing and preparing the previous disaster cases that would serve as the system's knowledge base. I helped with the identification of each category of the case and the important parts needed to describe each disaster situation.

**CHALLENGES:**

A challenge we both experienced was really making sure the system could properly identify the most similar disaster based on the given information. We did solve that by assigning weights to each factor, such as disaster type, flooding, area type, severity, and population, to calculate similarity scores. We learned how a system can use previous situations to help make decisions about a new event. We also learned about how similarity scores help find the most related case. It was an interesting task helping us learn more about CBR and its use in a programming or real-life scenario. Using previous cased, history, to predict a future action is very helpful.